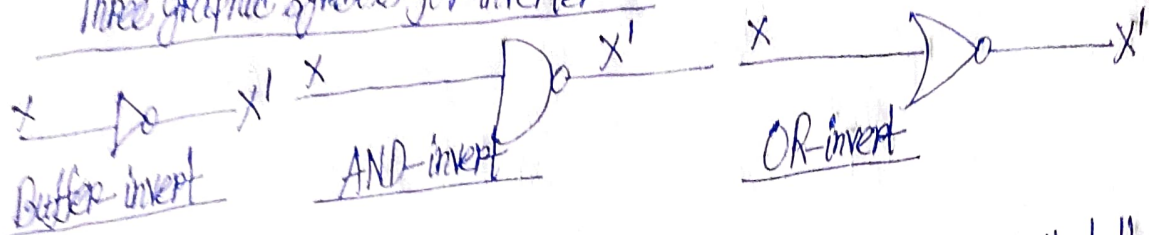


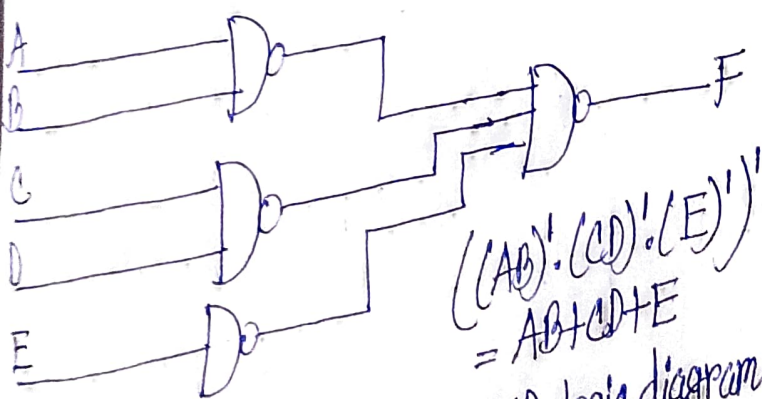
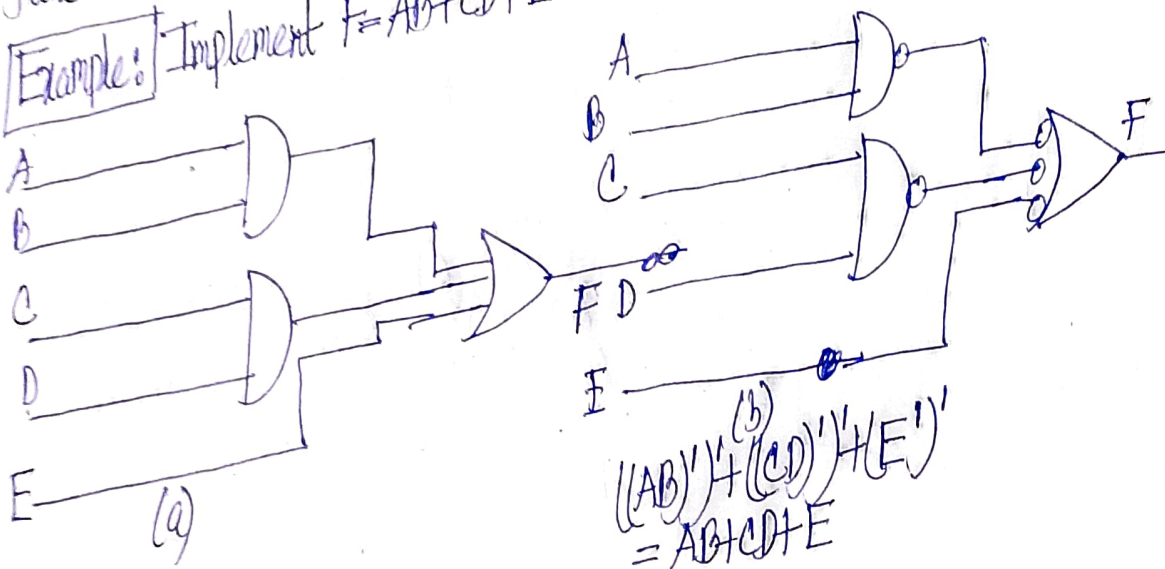
Three graphic symbols for inverter



3.6.1 NAND Implementation

The implementation of a Boolean function with NAND gates requires that the function be simplified in the sum of products form.

Example: Implement $F = AB + CD + E$



The rule for obtaining the NAND logic diagram from a Boolean function is as follows:

1. Simplify the function and express it in sum of products.
2. Draw a NAND gate for each product term of the function that has at least two literals. The inputs to each NAND gate, are the literals of the term. This constitutes a group of first-level gates.